WARDA SHAHZAD

PhD Chemistry: CZTS, Thin-Film, Photovoltaics, Semiconductor

CONTACT

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EDUCATION

PhD, Chemistry
COMSATS University Islamabad, Abbottabad Campus
2020 – 2026

MS, Chemistry
COMSATS University Islamabad, Abbottabad Campus
2017 – 2019

MSc, Chemistry
University of Wah, Pakistan
2015 – 2017

RESEARCH TOOLKIT

Materials Synthesis
Chemical bath deposition · Two-step thermal annealing · Composition screening

Structural Analysis
XRD · Williamson-Hall analysis · Crystallite size, microstrain & dislocation density · VESTA modeling

Composition & Morphology
SEM · EDX/EDS · Grain size analysis

Optical Analysis
UV-Vis spectroscopy · Tauc analysis · Refractive index · Dielectric & conductivity modeling

Computational
DFT: structural optimization, DOS, band structure

REFERENCES

Prof. Dr. Bushra Ismail
PhD Supervisor, COMSATS University Islamabad, Abbottabad Campus
bushraismail@cuiatd.edu.pk

Prof. Dr. Asad Muhammad Khan
Co-Supervisor, COMSATS University Islamabad, Abbottabad Campus
amk@cuiatd.edu.pk

Prof. Dr. Rafaqat Ali Khan
Supervisory Committee Member, COMSATS University Islamabad, Abbottabad Campus
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01 About Me

PhD researcher in Chemistry at COMSATS University Islamabad, Abbottabad Campus, studying how alkali metal doping (Li, Na, K) reshapes the structure and optical behavior of Cu_2ZnSnS_4 (CZTS) kesterite thin films for thin-film solar cells. My work targets suppression of Cu/Zn antisite defects and secondary phases (ZnS, Cu_2SnS_3), the primary loss mechanisms limiting CZTS device efficiency.

I synthesize films by chemical bath deposition and characterize them across XRD, SEM/EDX, and UV-Vis spectroscopy, using quantitative structural and optical proxies to track defect suppression across each doping series. My PhD thesis has been submitted for evaluation and I am currently awaiting the final viva voce. I am seeking postdoctoral opportunities to extend this work to new synthesis platforms and semiconductor material systems.

02 Research Experience

PhD Researcher, Chemistry *2020 – 2026*

COMSATS University Islamabad, Abbottabad Campus

- Investigated Li, Na, and K alkali-cation doping in CZTS kesterite thin films for defect engineering and secondary-phase suppression, using a systematic three-metal comparison on the same platform.
- Developed a two-step thermal annealing protocol for Li-doped CZTS that separated dopant effects from thermal effects, reducing dislocation density from 7.3 to 2.13×10^{14} m^{-2} and tuning bandgap from 1.36 to 1.27 eV.
- Showed Na doping narrows optical bandgap from 1.5 to 1.0 eV while reducing microstrain from 0.37 to 0.13×10^{-3}.
- Demonstrated K doping increases crystallite growth from 40 to 52 nm alongside a bandgap shift from 1.50 to 1.42 eV, while retaining the tetragonal kesterite structure.
- Presented research at SSCS Conference 2025 (2nd Prize, poster), ICRA-C 2024 (NUST Islamabad, oral), and ICC RTC&RET 2023 (University of Swat, poster); completed advanced analytical instrumentation training at the National Centre for Physics, Islamabad (2026).
- Mentored MS students; peer reviewer for Journal of Materials Science: Materials in Electronics (Springer Nature).

MS Researcher, Chemistry *2017 – 2019*

COMSATS University Islamabad, Abbottabad Campus

- Studied visible-light photocatalysis using alkaline-earth zirconate perovskite and graphene-oxide composites for wastewater treatment, introducing bandgap engineering concepts later applied to CZTS.

03 Publications

FIRST AUTHOR · PHD RESEARCH
Novel Insights from XRD and Photophysical Properties of Cu_2ZnSnS_4: Na Thin Films Deposited by Aqueous Chemical Route
Journal of Physics and Chemistry of Solids, Vol. 208, 113116 (2026)
DOI: 10.1016/j.jpcs.2025.113116

FIRST AUTHOR · PHD RESEARCH
Investigation of the Optical Properties of Solution Processed Rare Metal-free Lithium Doped Kesterite Thin Films
Optical Materials, Vol. 170, 117582 (2026)
DOI: 10.1016/j.optmat.2025.117582

FIRST AUTHOR · MS RESEARCH
Enhanced Visible Light Photocatalytic Performance of $Sr_{0.3}(Ba,Mn)_{0.7}ZrO_3$ Perovskites Anchored on Graphene Oxide
Ceramics International, Vol. 48, Issue 17, 24979–24988 (2022)
DOI: 10.1016/j.ceramint.2022.05.151

CO-AUTHOR
Transformation of Refractory Ceramic $MgAl_2O_4$ into Blue Light Emitting Nanomaterials by Sr^{2+}/Cr^{3+} Activation
Materials Science and Engineering: B, Vol. 303, 117273 (2024)
DOI: 10.1016/j.mseb.2024.117273

CO-AUTHOR
Assessment of Thermodynamic Properties of $SrSnO_3$ Perovskite for Enhanced Photocatalytic Applications
Journal of Inorganic and Organometallic Polymers and Materials, Vol. 35, 5767–5789 (2025)
DOI: 10.1007/s10904-025-03621-x